Linear Algebra. Spring 2023
Part I

1. Vector space axioms. Basic corollaries from axioms.
2. Subspaces. Linear combinations and span. Examples of subspaces in the polynomial vector space and the vector space of matrices.
3. Sum and intersection of the subspaces. Decomposition into a direct sum.
4. Linear dependence and independence. Basic lemmas.
5. $*$ Theorem o linear dependence of linear combinations (Theorem 3.4).
6. Basis. Coordinates of a vector relative to a basis. Examples of finding coordinates relative to a given basis.
7. Extending linear independent sets to a basis. Maximal linear independent set.
8. Extracting the basis from the spanning set. Minimal spanning set.
9. Dimension of the vector space. Dimension of the vector space of matrices and of the vector space of polynomials of restricted degree. Criteria for a subspace to coincide with ambient space.
10. Criteria for invertibility of transition matrix between a basis an some set ov vectors (Proposition 4.4).
11. Theorem on dimensions of the sum and the intersection.
12. $*$ Criteria of decomposition into a direct sum (Proposition 4.6).
13. Linear transformations. Geometric examples. Linear transformation is uniwuely defines by its value on some basis.
14. Sum of two linear transformations. The vector space $\mathcal{L}(V, W)$.
15. $*$ Composition of linear map. Both distributivity properties of the composition.
16. Kernel and the mage of a linear transformation are subspaces. Theorem on dimension of the kernel and the image. Application to a homogemeous linear system.
17. A matrix of a linear map relative to a pair of bases. Addition of matrices and of linear maps.
18. $*$ Composition of linear maps and matrix multiplication.
19. Isomorphism between a finite dimensional vector space and the coordinate space (Theorems 6.3 and 7.3).
20. $*$ Isomorphism between the space of linear map and the space of matrices.
21. Computation of the value of linear map using its matrix (Theorem 6.4).
22. Properties of the transistion matrix.
23. Relation between the matrices of the same linear map relative to different pair of bases (Theorem 6.7).
24. Canonical form of the linear map between two different spaces. Rank and the dimension of the image.
25. Criteria for invertibility of a linear operator (Proposition 7.7).
26. Eigenvectors of a linear operator and eigencolumns of its matrix.
27. Linear independence of eigenvectors and direct sum of eigenspaces.
28. Invariant subspaces. Restriction of an operator on its invariant subspace. Matrix of an operator having invariant subsoace relative to the basis chosen coherently with this subspace.
29. Characteristic polynomial. Independence of the choice of basis.
30. Eigenvalues are the roots of characteristic polynomial.
31. Trace of linear map. Relation to eigenvalues.
32. Polynomials in linear operator. $\mathrm{Ker}(p(L))$ and $\mathrm{Im}(p(L)$ are L-invariant subspace.
33. $*$ Hamilton-Caley theorem.
34. Criteria for diagonalizability.
35. Equivalent conditions for an operator to be nilpotent.
36. Upper triangular form fro the marix of a nilpotent operator.
37. Generalized eigensvectors and generalized eigenspace.

38. Decomposition into a direct sum of generalized eigenspace.
39. Jordan chains for a nilpotent operator. ∗ Construction of the basis consisting of Kordan chains of the given nilpotent operator.
40. Jordan block corresponding to a Jordan chain. Jordan normal form (Theorem 12.1).
41. Number of Jordan blocks corresponding to a given eigenvalue.
42. Jordan normal form of a matrix.
43. Finding the Jordan normal form of a matrix of the small size.
44. Exponentiation of a matrix usin Jordan normal form.
45. Square root extraction using Jordan normal form.

Part II

1. Lagrange method for a diagonalization od quadratic forms.
2. Definition of a bilinear form and its Gram matrix. Computation of the value of bilinear form in coordinates.
3. Relation between symmetric bilinear form and corresponding quadratic form.
4. Transformation of the Gram matrix under change of coordinates.
5. Orthogonal operators on a vector space equipped with a quadratic form.
6. Orthogonal complement and its basic properties in general cases (without non-degeneracy assumption).
7. Non-degenerate bilinear form. Equivalent conditions.
8. Orthogonal completion for a non-degenerate bilinear form.
9. Gram matrix of the bilinear form restricted on a subspace. The orthogonal completeion of the subspace such that the restriction is non-degenerate (Proposition 14.6).
10. Using orthogonal completion for finding the intersection of subspaces.
11. Equivalent conditions for real matrix to be orthogonal.
12. ∗ Riesz representation theorem.
13. Gram-Schmidt orthogonalization process.
14. Finding decomposition $A = LQ$ where L is low triangular and Q is orthogonal.
15. Orthogonal polynomial with respect to certain inner products.
16. Orthogonal direct sum decomposition. Orthogonal projector and its properties.
17. Formula for orthogonal projector using orthogonal basis. Fourier decomposition relative to an orthonormal basis.
18. Formula for an orthogonal projector onto some subspace in $\mathbb{R}^n$.
19. Characterisation of orthogonal projectors.
20. Euclidean space. Cauchy-Bunyakovsky and triangle inequality in the Euclidean space.
21. Minimal distance from a poit to a subspace in the Euclidean space.
22. Least square solution of the (inconsistent) linear system.
23. ∗ Interpolation problem and Lagrange interpolation formula.
24. Hermitian form on a complex vector space and its Gram matrix. Transformation of the Gram matrix under linear change of variables.
25. Unitary matrices.
26. Adjoint operator. Its existence and basic properties.
27. Eigenvalues and eigenvectors of the selfadjpint operator.
28. Examples of differential selfadjoint operators.
29. Canonical form of the selfadjoint operator. Application to diagonalization of a quadratic form by orthogonal change pf variables.

30. Normal operator on an inner space. Isometries are normal operators.
31. Canonical form of a normal operator on a complex inner space.
32. $*$ Canonical form of a normal operator on a Euclidean space.
33. Eigenvalues of orthogonal and unitary matrices.
34. Canonical form of isometry on Euclidean space.
35. Euler rotation theorem. Dectecting a type of an isometry on $\mathbb{R}^3$.
36. Reflections. Relation to othogonal projectors. Formula for a reflection along a vector.
37. Positive selfadjoint operators on an inner space.
38. Square root of the positive selfadjoint operator. $*$ Uniqueness.
39. The image of the unit sphere under linear transformation
40. Polar decomposition.
41. Reducing arbitrary operator on a complex inner space to a triangular form.
42. Singular values an singular value decompositon.
43. Operator norm of alinear map between two inner spaces. Its basic properties.
44. Operator norm is equal to the greatest singular value.
45. $*$ Relative error in measurement and it behaves under linear transformation. Condition number of a square ivertible marix.
46. Best approximation for a given linear map by a linear map of prescribed small rank.
47. Angle between a vector and a subspace in the Euclidean space. Minimum angle between two subspaces.
48. $*$ Angles between two planes in $\mathbb{R}^4$.